Inverse Variation and the Reciprocal Function

Name _____

Algebra 2 Lesson 4-1 (fast track) Date _____ Period _____

Learning goal: I can write an inverse variation equation from one pair of values and use it to find a missing value in a table or a word problem.

Language goal: I can explain how I found a missing value using the words *varies inversely*, *product*, *constant of variation*, and *inverse variation*.

Spanish: *variación inversa* (inverse variation), *constante de proporcionalidad* (constant of variation), *recíproco* (reciprocal).

On the quiz: You will (1) find a missing value in an inverse-variation table, (2) choose the equation $y = \frac{k}{x}$ that fits a table, and (3) solve a word problem where one quantity varies inversely with another. Example 3 gives you the two steps for all three: find k from a pair you know, then use k. Items (4) and (5), asymptotes and translations of $y = \frac{1}{x}$, are on the full Savvas quiz only.

Key idea. An **inverse variation** is modeled by $y = \frac{k}{x}$, where $k \neq 0$, so $xy = k$. The **product** of every pair is the same number k, the **constant of variation**. For positive x and y, as one increases, the other decreases.

x	1	2	3	4	6	8	12	24
y	24	12	8	6	4	3	2	1
$k = xy$	24	24	24	24	24	24	24	24

When x doubles (1, 2, 4, 8), y is halved (24, 12, 6, 3).

Example 3: Use an Inverse Variation Model

On a Greek bouzouki, the string length s varies inversely with the frequency f of its vibrations. (Players can vary string length by placing their fingers behind the frets.) The frequency of a 26-inch E-string is 329.63 cycles per second. What is the frequency when the string length is 13 inches?

Step 1 Write the equation for an inverse variation and solve for k (26 for s, 329.63 for f).

$$s = \frac{k}{f} \quad \longrightarrow \quad 26 = \frac{k}{329.63} \quad \longrightarrow \quad 26(329.63) = k \quad \longrightarrow \quad \mathbf{8{,}570.38 = k}$$

Step 2 Substitute $k = 8{,}570.38$ in the inverse variation equation and find f.

$$13 = \frac{8{,}570.38}{f} \quad \longrightarrow \quad 13f = 8{,}570.38 \quad \longrightarrow \quad f = \frac{8{,}570.38}{13} = \mathbf{659.26}$$

Answer: the 13-inch string has a frequency of **659.26 cycles per second**. **Check:** $13(659.26) = 8{,}570.38$, the same product. ✓ Half the length, double the frequency.

s (in.)	f (cycles/s)	$k = sf$
26	329.63	8,570.38
13	**659.26**	8,570.38

Say it: The string length *varies inversely* with the frequency, so the *product* sf is always _____. That number is the *constant of variation*, k. When the length is cut in half, the frequency _____ because

Try It 3: Use an inverse variation model

The amount of time it takes for an ice cube to melt varies inversely to the air temperature in degrees Celsius. At 20° Celsius, the ice will melt in 20 minutes. How long will it take the ice to melt if the temperature is 30° Celsius?

Step 1 Let $t =$ minutes and $d =$ degrees Celsius. Write $t = \frac{k}{d}$, substitute the pair you know, and solve for k.

$k = t \cdot d =$ _____ $\cdot$ _____ $=$ _____

Step 2 Substitute k and the new temperature. Find t.

$t =$ _____ $\div$ _____ $\approx$ _____ minutes

Check: multiply your new pair, using the unrounded time (a fraction), not the rounded one. Is the product the same k? _____

Does it make sense that warmer air gives a *shorter* time? Use the words *varies inversely.*

Practice: An inverse variation in a table

The table represents an inverse variation.

x	-5	-2	4	10
y	4	10	q	-2
xy	_____	_____		_____

a. Find the product xy for each column you know. The constant of variation is $k =$ _____.

b. Write the equation of the inverse variation: $y =$ _____

c. Find the value of q. Show your work.

d. Check: multiply 4 and your q. Do you get k? _____
One of x, y is negative in every pair because the *product* $xy = k$ is _____.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
C9D	Classify variation (warm-up)	Warm-up: a constant product xy means inverse variation
PNY ★	Write and solve inverse variation equations	Quiz 1, 2 and 3: find k, then use it (Example 3)
PXE ★	Find the constant of variation	Quiz 1 and 2: $k = xy$ from a table
ULD	Domain and range of reciprocal functions (optional)	Quiz 4 and 5 (full Savvas quiz only): asymptotes and translations of $y = 1/(x - h) + k$; not in this packet

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Model with mathematics

The time t required to empty a water tank varies inversely as the rate of pumping p. A pump can empty a water tank in 40 min at the rate of 120 gal/min. Write the equation of the inverse variation. How long will it take the pump to empty the water tank at the rate of 200 gal/min?

Equation: $t =$ _____ Time at 200 gal/min: _____ min

Because t *varies inversely* as p, the *product* _____ is the *constant of variation*, $k =$ _____.
A faster pump means _____ time, because ...

Before the quiz, finish these sentences.

In an *inverse variation*, the *product* xy is _____ for every pair.

To find a missing value, first find _____, then _____.